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November 5, 1999

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DEC 01 1999

Division of Minerals & Geology

Mr. Paul Banks
Banks and Gesso, LLC
720 Kipling St., Suite 117
Lakewood, Colorado 80215

Subject: Additional HELP Modeling for CKD Disposal Site

Dear Mr. Banks:

Grant Environmental, Inc. (Grant) is pleased to provide this summary of the results of the additional HELP modeling performed for the Southdown CKD disposal site. The HELP model input and output is included in the enclosed attachment.

As requested by the Colorado Division of Minerals and Geology in their letter of October 18, 1999, Grant performed an additional HELP model run to simulate an open, partially-filled CKD disposal site without an interim compacted shale cap. The open C-Pit disposal site was simulated as a partially-filled, 7-acre disposal area with approximately 20 feet of disposed CKD. This is the approximate amount of CKD expected to be accumulated in a 3 to 5 year period.

The CKD disposal site was simulated as an open landfill site, therefore precipitation run-off was not allowed to occur. Under this conservative model scenario, all precipitation that falls on the disposed CKD is either lost through natural evaporation or percolates through the disposed CKD. The default HELP model soil type used to simulate the disposed CKD (material texture 10) was the same as that used in previously submitted HELP model runs. Default evapotranspiration and precipitation data for Denver, Colorado were also used in the simulation.

The results of this additional HELP model run indicate that the predicted flux of leachate through the disposed CKD in an open disposal site without an interim shale cap is about 0.34 inches per year. For the 7-acre disposal site, the predicted leachate flux is equivalent to about 8,581 cubic feet of leachate per year. The predicted leachate flux for the open, uncapped CKD disposal site is essentially the same as the leachate flux predicted for an open CKD disposal site with an interim shale cap. A summary of the annual leachate fluxes generated under the various C-Pit disposal site configurations previously modeled and the results of this analysis is provided in Table 1.

Table 1
Summary of the Predicted Leachate Fluxes for Various C-Pit Disposal Site Configurations For an Equivalent 7-Acre Disposal Site Area

C-Pit Configuration	Annual Leachate Flux		
	(inches)	(cubic feet)	
Open, No Cap	0.34	8,581	3.96 x 10^-5
Open, Interim Cap	0.30	7,592	3.95 x 10^-5
Closed, Final Cap	0.74	18,803	3.93 x 10^-5
Closed, Interim/Final Cap	0.04	1,131	3.53 x 10^-5

(303) 790-7400

The results of this and previously performed HELP model analyses of the CKD disposal pit suggest that the quantity of leachate predicted to be generated, in conjunction with the hydrogeologic conditions present at the C-Pit, and the physical and chemical characteristics of the CKD, do not pose a significant threat to shallow groundwater. The HELP modeling results suggests that proposed interim/final shale capping of the disposed CKD will provide the most protection of existing and potential future shallow groundwater use by reducing the quantity of leachate formed to an insignificant amount.

Grant appreciates the opportunity to work with you on this project. If you have any questions concerning this transmittal, please feel free to call me.

Sincerely,

Grant Environmental, Inc.

Richard L. Henry, P.G.
Chief Hydrogeologist

Gary N. Cantrell, P.E.
Senior Engineer

Attachment: As stated

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**                                                                          **
**                                                                          **
**              HYDROLOGIC EVALUATION OF LANDFILL PERFORMANCE               **
**                HELP MODEL VERSION 3.07  (1 NOVEMBER 1997)                **
**                  DEVELOPED BY ENVIRONMENTAL LABORATORY                   **
**                    USAE WATERWAYS EXPERIMENT STATION                     **
**             FOR USEPA RISK REDUCTION ENGINEERING LABORATORY              **
**                                                                          **
**                                                                          **
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PRECIPITATION DATA FILE:     D:\HELP3\DATA4.D4
TEMPERATURE DATA FILE:       D:\HELP3\DATA7.D7
SOLAR RADIATION DATA FILE:   D:\HELP3\DATA13.D13
EVAPOTRANSPIRATION DATA:     D:\HELP3\DATA11.D11
SOIL AND DESIGN DATA FILE:   D:\HELP3\OPEN5.D10
OUTPUT DATA FILE:            D:\HELP3\OPEN5.OUT


TIME:  19:20     DATE:  11/ 4/1999


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     TITLE:  OPEN C-PIT, 7 ACRE AREA, HALF FULL

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     NOTE:  INITIAL MOISTURE CONTENT OF THE LAYERS AND SNOW WATER WERE
              COMPUTED AS NEARLY STEADY-STATE VALUES BY THE PROGRAM.


                                   LAYER  1
                                   --------

                     TYPE 1 - VERTICAL PERCOLATION LAYER
                          MATERIAL TEXTURE NUMBER  10
          THICKNESS                 =   240.00  INCHES     20' CKD
          POROSITY                  =     0.3980 VOL/VOL
          FIELD CAPACITY            =     0.2440 VOL/VOL
          WILTING POINT             =     0.1360 VOL/VOL
          INITIAL SOIL WATER CONTENT =    0.2406 VOL/VOL
          EFFECTIVE SAT. HYD. COND.  = 0.119999997000E-03 CM/SEC   * - Conservative K
        NOTE:  SATURATED HYDRAULIC CONDUCTIVITY IS MULTIPLIED BY  1.34      2x10 4
                 FOR ROOT CHANNELS IN TOP HALF OF EVAPORATIVE ZONE.
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GENERAL DESIGN AND EVAPORATIVE ZONE DATA

NOTE: SCS RUNOFF CURVE NUMBER WAS COMPUTED FROM DEFAULT SOIL DATA BASE USING SOIL TEXTURE #10 WITH BARE GROUND CONDITIONS, A SURFACE SLOPE OF 2.% AND A SLOPE LENGTH OF 300. FEET.

SCS RUNOFF CURVE NUMBER	=	93.90		
FRACTION OF AREA ALLOWING RUNOFF	=	0.0	PERCENT	
AREA PROJECTED ON HORIZONTAL PLANE	=	7.000	ACRES	
EVAPORATIVE ZONE DEPTH	=	14.0	INCHES	1'2"
INITIAL WATER IN EVAPORATIVE ZONE	=	2.606	INCHES	
UPPER LIMIT OF EVAPORATIVE STORAGE	=	5.572	INCHES	
LOWER LIMIT OF EVAPORATIVE STORAGE	=	1.904	INCHES	
INITIAL SNOW WATER	=	0.000	INCHES	
INITIAL WATER IN LAYER MATERIALS	=	57.749	INCHES	almost 5'
TOTAL INITIAL WATER	=	57.749	INCHES	
✓ TOTAL SUBSURFACE INFLOW	=	0.00	INCHES/YEAR	– vadose zone

EVAPOTRANSPIRATION AND WEATHER DATA

NOTE: EVAPOTRANSPIRATION DATA WAS OBTAINED FROM DENVER COLORADO

STATION LATITUDE	=	39.77	DEGREES
MAXIMUM LEAF AREA INDEX	=	0.50	
START OF GROWING SEASON (JULIAN DATE)	=	122	
END OF GROWING SEASON (JULIAN DATE)	=	286	
EVAPORATIVE ZONE DEPTH	=	14.0	INCHES
AVERAGE ANNUAL WIND SPEED	=	8.80	MPH
AVERAGE 1ST QUARTER RELATIVE HUMIDITY	=	54.00	%
AVERAGE 2ND QUARTER RELATIVE HUMIDITY	=	50.00	%
AVERAGE 3RD QUARTER RELATIVE HUMIDITY	=	49.00	%
AVERAGE 4TH QUARTER RELATIVE HUMIDITY	=	54.00	%

NOTE: PRECIPITATION DATA FOR DENVER COLORADO WAS ENTERED FROM THE DEFAULT DATA FILE.

NOTE: TEMPERATURE DATA WAS SYNTHETICALLY GENERATED USING COEFFICIENTS FOR DENVER COLORADO

NORMAL MEAN MONTHLY TEMPERATURE (DEGREES FAHRENHEIT)

JAN/JUL	FEB/AUG	MAR/SEP	APR/OCT	MAY/NOV	JUN/DEC

29.50	33.60	38.00	47.40	57.20	67.00
73.30	71.40	62.60	51.90	38.70	32.60

NOTE: SOLAR RADIATION DATA WAS SYNTHETICALLY GENERATED USING COEFFICIENTS FOR DENVER COLORADO AND STATION LATITUDE = 39.77 DEGREES

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MONTHLY TOTALS (IN INCHES) FOR YEAR 1974

	JAN/JUL	FEB/AUG	MAR/SEP	APR/OCT	MAY/NOV	JUN/DEC
PRECIPITATION	1.03	0.82	1.32	2.28	0.06	2.01
	2.34	0.16	0.98	1.68	1.06	0.29
RUNOFF	0.000	0.000	0.000	0.000	0.000	0.000
	0.000	0.000	0.000	0.000	0.000	0.000
EVAPOTRANSPIRATION	1.087	0.833	1.406	2.183	0.630	1.941
	1.344	1.069	0.499	1.587	0.910	0.409
PERCOLATION/LEAKAGE THROUGH	0.0000	0.0000	0.0000	0.0000	0.0000	0.0071
LAYER 1	0.0911	0.0443	0.0000	0.0000	0.0000	0.0028

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ANNUAL TOTALS FOR YEAR 1974

	INCHES	CU. FEET	PERCENT
PRECIPITATION	14.03	356502.469	100.00
RUNOFF	0.000	0.000	0.00
EVAPOTRANSPIRATION	13.899	353166.531	99.06
PERC./LEAKAGE THROUGH LAYER 1	0.145161	3688.532	1.03
CHANGE IN WATER STORAGE	-0.014	-352.637	-0.10
SOIL WATER AT START OF YEAR	57.749	1467398.620	

SOIL WATER AT END OF YEAR	57.735	1467046.000	
SNOW WATER AT START OF YEAR	0.000	0.000	0.00
SNOW WATER AT END OF YEAR	0.000	0.000	0.00
ANNUAL WATER BUDGET BALANCE	0.0000	0.025	0.00

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MONTHLY TOTALS (IN INCHES) FOR YEAR 1975

	JAN/JUL	FEB/AUG	MAR/SEP	APR/OCT	MAY/NOV	JUN/DEC
PRECIPITATION	0.23	0.37	1.19	1.14	2.80	2.11
	2.78	2.00	0.24	0.30	1.88	0.47
RUNOFF	0.000	0.000	0.000	0.000	0.000	0.000
	0.000	0.000	0.000	0.000	0.000	0.000
EVAPOTRANSPIRATION	0.326	0.403	0.545	1.495	1.890	3.436
	2.387	2.661	0.184	0.130	0.952	0.499
PERCOLATION/LEAKAGE THROUGH	0.0000	0.0000	0.0000	0.0000	0.0000	0.0131
LAYER 1	0.1182	0.0330	0.0000	0.0000	0.0000	0.0000

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ANNUAL TOTALS FOR YEAR 1975

	INCHES	CU. FEET	PERCENT
PRECIPITATION	15.51	394109.250	100.00
RUNOFF	0.000	0.000	0.00
EVAPOTRANSPIRATION	14.908	378806.219	96.12
PERC./LEAKAGE THROUGH LAYER 1	0.164276	4174.251	1.06
CHANGE IN WATER STORAGE	0.438	11128.507	2.82
SOIL WATER AT START OF YEAR	57.735	1467046.000	

SOIL WATER AT END OF YEAR	58.173	1478174.500	
SNOW WATER AT START OF YEAR	0.000	0.000	0.00
SNOW WATER AT END OF YEAR	0.000	0.000	0.00
ANNUAL WATER BUDGET BALANCE	0.0000	0.272	0.00

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MONTHLY TOTALS (IN INCHES) FOR YEAR 1976

	JAN/JUL	FEB/AUG	MAR/SEP	APR/OCT	MAY/NOV	JUN/DEC
PRECIPITATION	0.19	0.54	1.34	1.27	1.34	0.63
	2.31	2.50	1.88	0.93	0.32	0.16
RUNOFF	0.000	0.000	0.000	0.000	0.000	0.000
	0.000	0.000	0.000	0.000	0.000	0.000
EVAPOTRANSPIRATION	0.321	0.674	0.847	1.010	1.758	0.941
	2.939	2.644	0.980	1.060	0.481	0.159
PERCOLATION/LEAKAGE THROUGH	0.0000	0.0000	0.0015	0.0016	0.0034	0.0032
LAYER 1	0.0000	0.0373	0.0241	0.0000	0.0000	0.0000

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ANNUAL TOTALS FOR YEAR 1976

	INCHES	CU. FEET	PERCENT
PRECIPITATION	13.41	340748.250	100.00
RUNOFF	0.000	0.000	0.00
EVAPOTRANSPIRATION	13.814	351021.469	103.01
PERC./LEAKAGE THROUGH LAYER 1	0.071240	1810.199	0.53
CHANGE IN WATER STORAGE	-0.476	-12083.669	-3.55
SOIL WATER AT START OF YEAR	58.173	1478174.500	

SOIL WATER AT END OF YEAR	57.697	1466090.870	
SNOW WATER AT START OF YEAR	0.000	0.000	0.00
SNOW WATER AT END OF YEAR	0.000	0.000	0.00
ANNUAL WATER BUDGET BALANCE	0.0000	0.238	0.00

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MONTHLY TOTALS (IN INCHES) FOR YEAR 1977

	JAN/JUL	FEB/AUG	MAR/SEP	APR/OCT	MAY/NOV	JUN/DEC
PRECIPITATION	0.16	0.27	1.24	2.13	0.34	1.02
	2.98	0.93	0.17	0.09	0.95	0.06
RUNOFF	0.000	0.000	0.000	0.000	0.000	0.000
	0.000	0.000	0.000	0.000	0.000	0.000
EVAPOTRANSPIRATION	0.235	0.340	1.144	2.358	0.648	0.416
	3.012	1.046	0.422	0.158	0.440	0.196
PERCOLATION/LEAKAGE THROUGH	0.0000	0.0000	0.0000	0.0020	0.0036	0.0000
LAYER 1	0.0038	0.0154	0.1374	0.0279	0.0000	0.0000

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ANNUAL TOTALS FOR YEAR 1977

	INCHES	CU. FEET	PERCENT
PRECIPITATION	10.34	262739.469	100.00
RUNOFF	0.000	0.000	0.00
EVAPOTRANSPIRATION	10.414	264631.344	100.72
PERC./LEAKAGE THROUGH LAYER 1	0.189978	4827.345	1.84
CHANGE IN WATER STORAGE	-0.264	-6719.289	-2.56
SOIL WATER AT START OF YEAR	57.697	1466090.870	

SOIL WATER AT END OF YEAR	57.433	1459371.500	
SNOW WATER AT START OF YEAR	0.000	0.000	0.00
SNOW WATER AT END OF YEAR	0.000	0.000	0.00
ANNUAL WATER BUDGET BALANCE	0.0000	0.084	0.00

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MONTHLY TOTALS (IN INCHES) FOR YEAR 1978

	JAN/JUL	FEB/AUG	MAR/SEP	APR/OCT	MAY/NOV	JUN/DEC
PRECIPITATION	0.27	0.27	1.07	1.42	3.86	1.17
	0.54	0.26	0.07	1.45	0.50	0.79
RUNOFF	0.000	0.000	0.000	0.000	0.000	0.000
	0.000	0.000	0.000	0.000	0.000	0.000
EVAPOTRANSPIRATION	0.297	0.345	0.614	1.480	3.002	1.281
	0.576	0.509	0.075	0.462	1.139	0.679
PERCOLATION/LEAKAGE THROUGH	0.0000	0.0000	0.0000	0.0000	0.0066	0.0401
LAYER 1	0.1569	0.2228	0.2106	0.2056	0.2181	0.0573

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ANNUAL TOTALS FOR YEAR 1978

	INCHES	CU. FEET	PERCENT
PRECIPITATION	11.67	296534.781	100.00
RUNOFF	0.000	0.000	0.00
EVAPOTRANSPIRATION	10.461	265812.625	89.64
PERC./LEAKAGE THROUGH LAYER 1	1.117910	28406.084	9.58
CHANGE IN WATER STORAGE	0.091	2316.080	0.78
SOIL WATER AT START OF YEAR	57.433	1459371.500	

SOIL WATER AT END OF YEAR	57.524	1461687.620	
SNOW WATER AT START OF YEAR	0.000	0.000	0.00
SNOW WATER AT END OF YEAR	0.000	0.000	0.00
ANNUAL WATER BUDGET BALANCE	0.0000	-0.030	0.00

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AVERAGE MONTHLY VALUES IN INCHES FOR YEARS 1974 THROUGH 1978

	JAN/JUL	FEB/AUG	MAR/SEP	APR/OCT	MAY/NOV	JUN/DEC
PRECIPITATION						
TOTALS	0.38	0.45	1.23	1.65	1.68	1.39
	2.19	1.17	0.67	0.89	0.94	0.35
STD. DEVIATIONS	0.37	0.23	0.11	0.52	1.62	0.65
	0.97	1.04	0.77	0.69	0.61	0.29
RUNOFF						
TOTALS	0.000	0.000	0.000	0.000	0.000	0.000
	0.000	0.000	0.000	0.000	0.000	0.000
STD. DEVIATIONS	0.000	0.000	0.000	0.000	0.000	0.000
	0.000	0.000	0.000	0.000	0.000	0.000
EVAPOTRANSPIRATION						
TOTALS	0.453	0.519	0.911	1.705	1.586	1.603
	2.051	1.586	0.432	0.679	0.785	0.389
STD. DEVIATIONS	0.356	0.223	0.363	0.555	0.990	1.165
	1.061	1.000	0.351	0.630	0.308	0.216
PERCOLATION/LEAKAGE THROUGH LAYER 1						
TOTALS	0.0000	0.0000	0.0003	0.0007	0.0027	0.0127
	0.0740	0.0706	0.0744	0.0467	0.0436	0.0120
STD. DEVIATIONS	0.0000	0.0000	0.0007	0.0010	0.0028	0.0161
	0.0699	0.0858	0.0950	0.0896	0.0975	0.0253

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AVERAGE ANNUAL TOTALS & (STD. DEVIATIONS) FOR YEARS 1974 THROUGH 1978

	INCHES		CU. FEET	PERCENT
PRECIPITATION	12.99	(2.023)	330126.9	100.00
RUNOFF	0.000	(0.0000)	0.00	0.000
EVAPOTRANSPIRATION	12.699	(2.1089)	322687.62	97.747
PERCOLATION/LEAKAGE THROUGH LAYER 1	0.33771	(0.43838)	8581.283	2.59939
CHANGE IN WATER STORAGE	-0.045	(0.3487)	-1142.20	-0.346

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PEAK DAILY VALUES FOR YEARS 1974 THROUGH 1978

	(INCHES)	(CU. FT.)
PRECIPITATION	1.79	45483.898
RUNOFF	0.000	0.0000
PERCOLATION/LEAKAGE THROUGH LAYER 1	0.011186	284.23792
SNOW WATER	1.19	30343.0996
MAXIMUM VEG. SOIL WATER (VOL/VOL)	0.2938	
MINIMUM VEG. SOIL WATER (VOL/VOL)	0.1360	

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FINAL WATER STORAGE AT END OF YEAR 1978

LAYER	(INCHES)	(VOL/VOL)
1	57.5241	0.2397

SNOW WATER 0.000

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